



ENGINEERING MATHEMATICS-II

Department of Science and Humanities

ENGINEERING MATHEMATICS-II

UNIT 1 : INTEGRAL CALCULUS

class 6:TRIPLE INTEGRAL

TRIPLE INTEGRAL



As the name implies, triple integrals are three successive integrations. They are used to calculate the Volume, or to integrate in a 4th dimension over three other independent dimensions

$$\iiint_V f(x, y, z) dV$$

A triple integral represents a summation in a hypothetical 4th dimension. To understand this, imagine a slightly different scenario, Where the first 3 dimensions are space, space, time and that the fourth dimension is the third space.

Let z be a function of x and y and in turn x and y be functions of 't'.

INTUITION BEHIND TRIPLE INTEGRAL



Integrating z with respect to x and y gives us the volume as a function of t . If we plug in a value of t , this gives us the numerical volume at a particular instance of time. Thus, if we integrate this result with respect to t we are adding up all the different volumes over a period.

For example, imagine a balloon that is being inflated. At any particular instant of time, we can use a double integral to calculate its volume. However, if we wanted to sum up all the volumes over the entire inflation process, we must use triple integral as described above.

Instead of a small interval or a small rectangle, there is a small box. Instead of length dx or area $dx dy$, the box has volume $dv = dx dy dz$

INTUITION BEHIND TRIPLE INTEGRAL

Consider the rectangular box

$$\begin{aligned} B &= \{(x, y, z) \in \mathbb{R}^3 : a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\} \\ &= [a, b] \times [c, d] \times [r, s] \end{aligned}$$

We proceed as follows:

1. Divide B into sub-boxes. To achieve this, we do the following:
 - (a) We divide $[a, b]$ into l subintervals $[x_{i-1}, x_i]$ of equal width Δx .
 - (b) We divide $[c, d]$ into m subintervals $[y_{j-1}, y_j]$ of equal width Δy .
 - (c) We divide $[r, s]$ into n subintervals $[z_{k-1}, z_k]$ of equal width Δz .
 - (d) Then, B is divided into lmn sub-boxes B_{ijk} Where,

INTUITION BEHIND TRIPLE INTEGRAL

$$B_{ijk} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k]$$

The volume of B_{ijk} is $\Delta V = \Delta x \Delta y \Delta z$.

2. In each sub-box B_{ijk} , we select a sample point $(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)$.
3. We form the triple Riemann sum

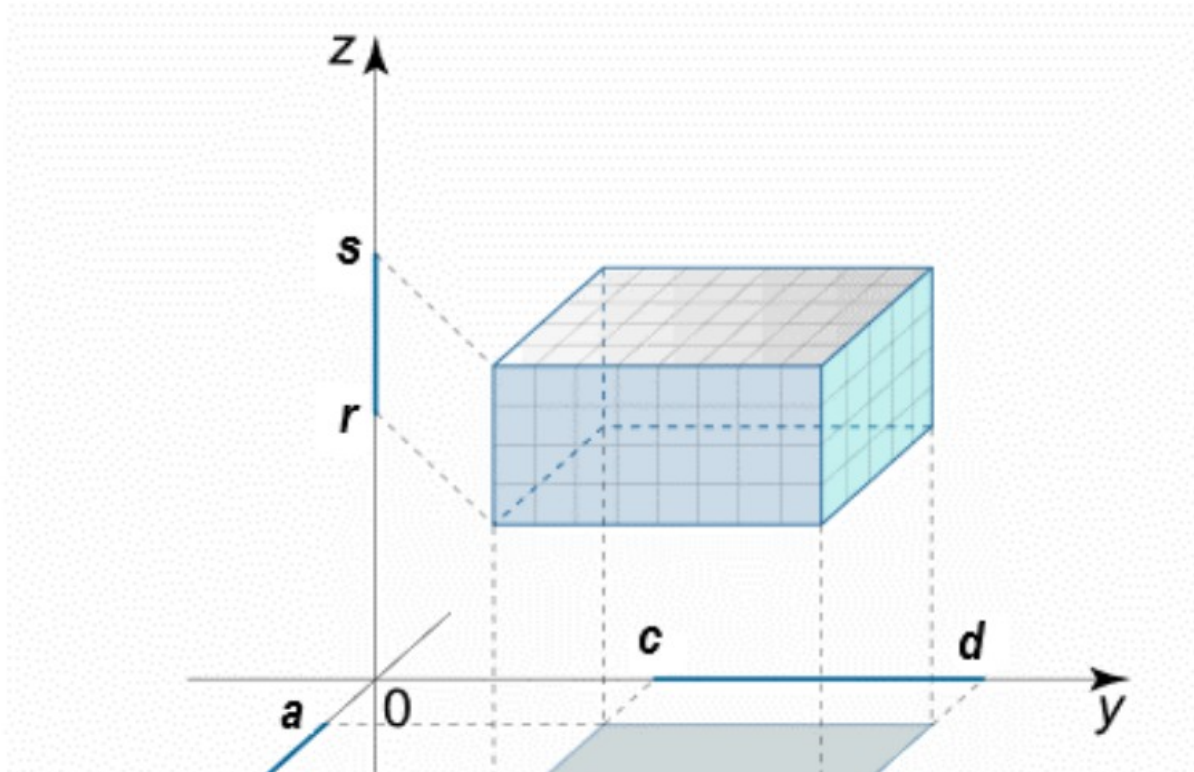
$$\sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

Definition of Triple Integral

Definition 4.7.1 The *triple integral* of f over the box B is

$$\iiint_B f(x, y, z) dV = \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

It can be shown that the triple integral always exists if f is continuous.



Definition of Triple Integral

If f is continuous over a bounded solid region Q , then the **triple integral of f over Q** is defined as

$$\iiint_Q f(x, y, z) dV = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta V_i$$

provided the limit exists. The **volume** of the solid region Q is given by

$$\text{Volume of } Q = \iiint_Q dV.$$

TRIPLE INTEGRAL OVER A CUBOID

If $f : [a, b] \times [c, d] \times [e, g] \rightarrow \mathbb{R}$ is a continuous function, then

$$\begin{aligned} \iiint_{[a,b] \times [c,d] \times [e,g]} f(x, y, z) \, dx \, dy \, dz &= \int_a^b \int_c^d \int_e^g f(x, y, z) \, dz \, dy \, dx \\ &= \int_a^b \int_e^g \int_c^d f(x, y, z) \, dy \, dz \, dx \\ &= \int_c^d \int_a^b \int_e^g f(x, y, z) \, dz \, dx \, dy \\ &= \int_c^d \int_e^g \int_a^b f(x, y, z) \, dx \, dz \, dy \\ &= \int_e^g \int_a^b \int_c^d f(x, y, z) \, dy \, dx \, dz \\ &= \int_e^g \int_c^d \int_a^b f(x, y, z) \, dx \, dy \, dz. \end{aligned}$$

EVALUATION OF TRIPLE INTEGRAL

Let f be continuous on a solid region Q defined by

$$a \leq x \leq b, \quad h_1(x) \leq y \leq h_2(x), \quad g_1(x, y) \leq z \leq g_2(x, y)$$

where h_1 , h_2 , g_1 , and g_2 are continuous functions. Then,

$$\iiint_O f(x, y, z) dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz dy dx.$$

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(y, z)}^{u_2(y, z)} f(x, y, z) dx \right] dA$$

EVALUATION OF TRIPLE INTEGRALS

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(y, z)}^{u_2(y, z)} f(x, y, z) dx \right] dA$$

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, z)}^{u_2(x, z)} f(x, y, z) dy \right] dA$$

EVALUATION OF TRIPLE INTEGRALS

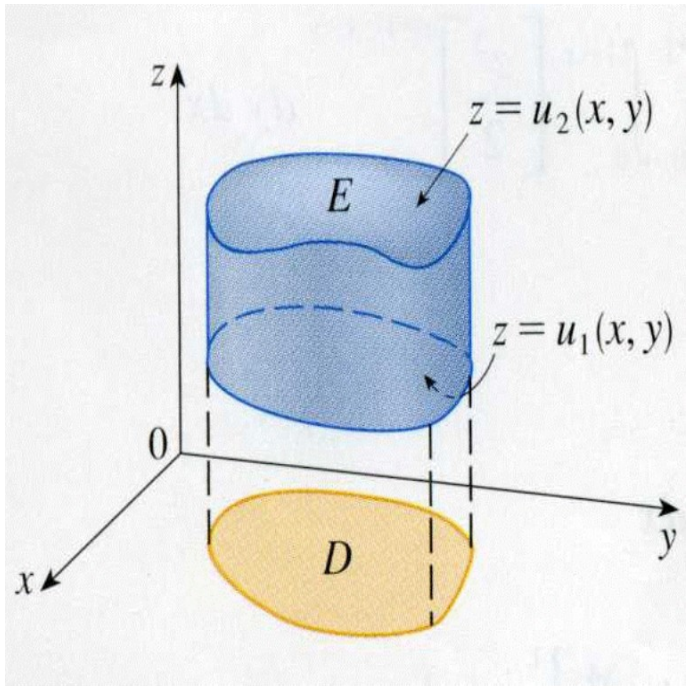
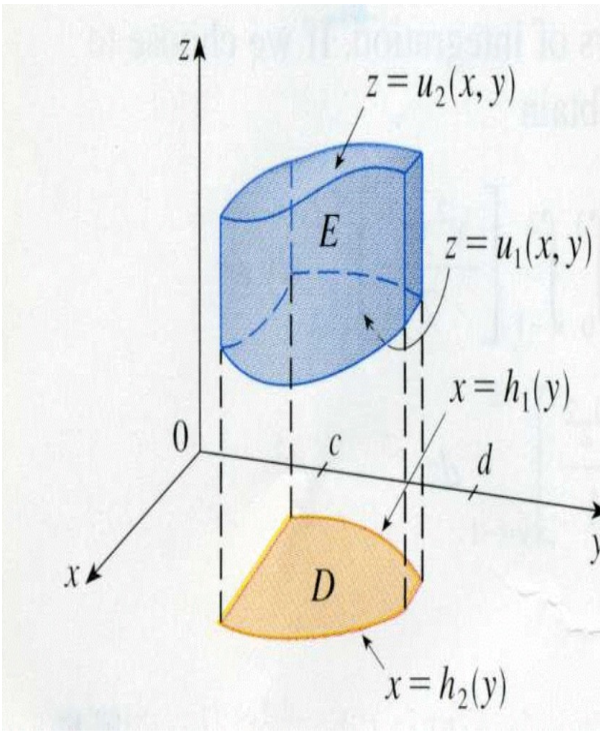
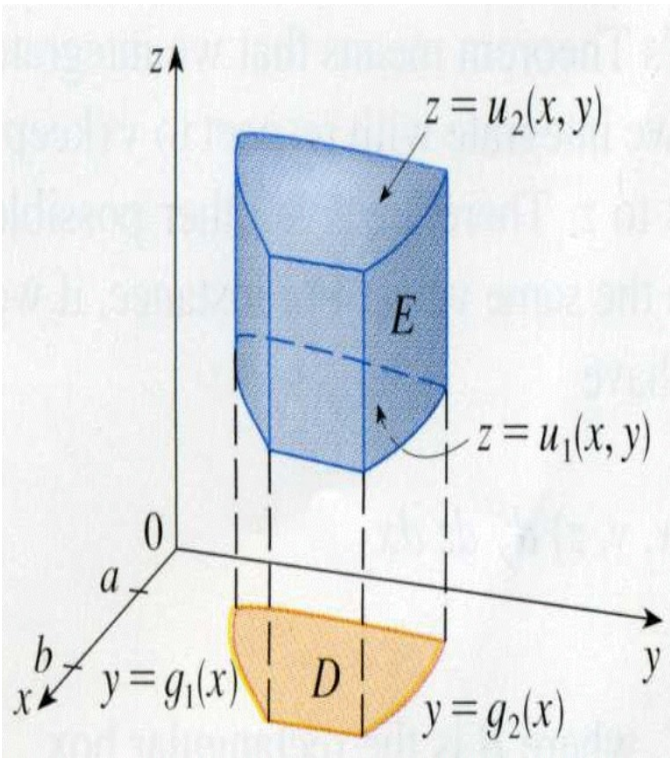


Figure 4.30: A Type 1 Solid



PROPERTIES OF TRIPLE INTEGRALS

Some of the properties of double integrals can be restated in terms of triple integrals.

$$1. \iiint_Q cf(x, y, z) dV = c \iiint_Q f(x, y, z) dV$$

$$2. \iiint_Q [f(x, y, z) \pm g(x, y, z)] dV = \iiint_Q f(x, y, z) dV \pm \iiint_Q g(x, y, z) dV$$

$$3. \iiint_Q f(x, y, z) dV = \iiint_{Q_1} f(x, y, z) dV + \iiint_{Q_2} f(x, y, z) dV$$

EXAMPLE1 : TRIPLE INTEGRAL

Evaluate $\int_0^2 \int_1^3 \int_1^2 xy^2 z dz dy dx$

Solution:

$$\begin{aligned}\int_0^2 \int_1^3 \int_1^2 xy^2 z dz dy dx &= \int_0^2 \int_1^3 \left[\frac{z^2}{2} \right]_1^2 xy^2 dy dx \\ &= \int_0^2 \int_1^3 \frac{3}{2} xy^2 dy dx \\ &= \frac{3}{2} \int_0^2 \left[\frac{y^3}{3} \right]_1^3 x dx \\ &= \frac{26}{2} \left[\frac{x^2}{2} \right]_0^2 = 26\end{aligned}$$

EXAMPLE 2 : TRIPLE INTEGRAL

Evaluate $\int_0^a \int_0^x \int_0^y xyz dx dy dz$

Solution:

$$\begin{aligned} I &= \int_0^a \int_0^x \left[\int_0^y z dz \right] xy dy dx \\ &= \int_0^a \int_0^x \left[\frac{z^2}{2} \right]_0^y xy dy dx \\ &= \int_0^a \int_0^x \left[\frac{y^2}{2} \right] xy dy dx \\ &= \int_0^a \int_0^x \left[\frac{y^3}{2} \right] dy x dx = \int_0^a \left[\frac{y^4}{8} \right]_0^x x dx \\ &= \left[\frac{x^6}{48} \right]_0^a = \frac{a^6}{48} \end{aligned}$$



THANK YOU
